Project Report: Cyclistic Bike-Share Analysis

1. Ask: Business Task

- **Business Task Statement:** The company believes their success relies on maximizing the number of annual memberships. Therefore, we need to understand how casual riders and annual members use Cyclistic bikes differently. Once we get an understanding of how, we then will be able to come up with a marketing strategy to convert casual members to annual members.
- **Primary Stakeholders:** Lily Moreno, Cyclistic marketing analytics team, Cyclistic executive team

2. Prepare: Data Sources

- **Data Description:** The data is Cyclistic's historical trip data, provided by Motivate International Inc., covering the previous 12 months of trip data.
- **Data Organization:** The data is structured as historical trip data, typically available in 12 monthly files.
- **Bias and Credibility (ROCCC):** This public dataset is considered appropriate and is Original, Reliable, Comprehensive, Current, and Cited for the case study.
- **Licensing and Privacy:** The data is made available by Motivate International Inc. under a specific license. Data-privacy issues prohibit the use of riders' personally identifiable information (PII), which has been removed to prevent connecting pass purchases to credit card numbers or determining riders' residency.

3. Process: Documentation of Cleaning

Tools Chosen

I chose to use **Python (with the Pandas library)** via a Jupyter Notebook environment for the cleaning and processing phase.

- **Scalability:** The raw data consists of 12 separate CSV files totaling over **5 million rows**. This volume exceeds the limits of standard spreadsheet software (Excel/Google Sheets), which would lead to performance degradation and potential data loss.
- **Reproducibility:** Using Python allows for a scripted pipeline. Any future data updates can be processed through the same logic with a single "Run All" command, ensuring consistency.
- **Complex Filtering:** Python's string manipulation and conditional filtering (RegEx) allowed for a more sophisticated removal of maintenance and test records than manual sorting would permit.

Cleaning Steps

Step 1: Data Ingestion & Consolidation

- Utilized the glob and os libraries to programmatically identify all 12 monthly CSV files in the data_raw directory.
- Merged the files into a single unified DataFrame (df_all_trips) using pd.concat, ensuring all 13 columns were aligned correctly across the different time periods.

Step 2: Feature Engineering (Time & Date)

- Converted the started_at and ended_at columns from objects to **datetime64[ns]** format to enable mathematical operations.
- **Calculated 'ride_length':** Created a new column by subtracting the start time from the end time. The results were converted to minutes to provide a human-readable metric for stakeholders.
- **Calculated 'day_of_week':** Extracted the day of the week from the started_at timestamp and mapped the integer results (0-6) to string names (e.g., "Monday").

Step 3: Data Validation & Scrubbing

- **Duplicate Removal:** Identified and dropped duplicate records based on the ride_id primary key to ensure each trip was unique.
- **Minimum Threshold Filter:** Filtered the dataset to only include trips with a ride_length of **1 minute or greater**. This removed thousands of "false starts" or docking errors where trip duration was less than 60 seconds.
- **NULL Handling:** Removed records missing critical end_lat or end_lng coordinates to ensure the data remained viable for future geospatial analysis.

Step 4: Logic-Based Maintenance Filtering

- Used string matching (Regular Expressions) to identify and remove rides associated with **test stations, maintenance hubs, or internal quality control checks** (e.g., station names containing "test," "trial," or "base").
- **Final Result:** Successfully cleaned and standardized the dataset for analysis. By filtering out system test data, null station entries, and extreme outliers, the final dataset was optimized for accurate behavioral modeling between members and casual riders. Final Result: Successfully cleaned and standardized the dataset for analysis. By filtering out system test data, null station entries, and extreme outliers, the final dataset was optimized for accurate behavioral modeling between members and casual riders.

Step 5: Categorical Formatting & Export

- Set a **categorical sort order** for the day_of_week column (Sunday through Saturday). This ensures that all downstream visualizations in the "Analyze" and "Share" phases are automatically sorted chronologically rather than alphabetically.
- Exported the final cleaned dataset as a CSV file to the data_processed folder for use in Tableau and the Analysis notebook.

4. Analyze: Summary of Findings

- Based on the analysis of 5,446,229 cleaned records, the following baseline metrics were established:
- **Mean (Average) Ride Length:**
    - **Casual Riders:** 19.97 minutes
    - **Annual Members:** 12.27 minutes
- **Max Ride Length:** 1,499.97 minutes (approximately 25 hours, indicating bikes kept overnight).
- **Mode (Most Frequent Day):**
    - **Casual Riders:** Saturday (397,315 rides)
    - **Annual Members:** Thursday (567,363 rides)
- **Usage Trends**
- The data reveals a distinct divergence in user behavior. **Annual Members** utilize the service as a functional utility for commuting, characterized by high volume and short, consistent durations. **Casual Riders** utilize the service for recreation and tourism,

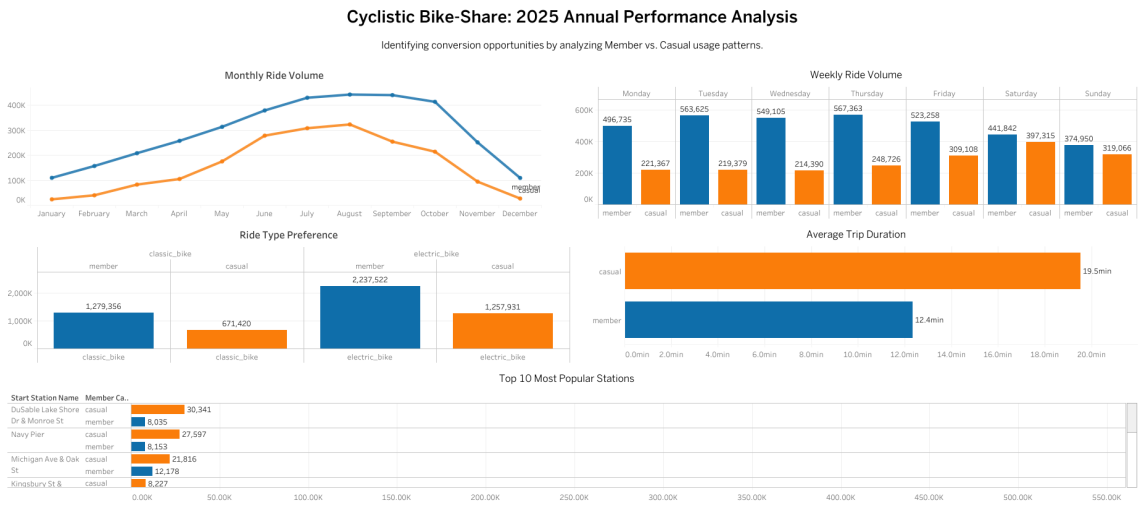
characterized by significant surges in duration and volume during the weekends and summer months.

- **Trend 1: Weekend Recreational Surge**
- Casual riders exhibit a "Leisure Profile." Their average trip duration increases by **41%** on Sundays compared to mid-week lows (increasing from 16.47 to 23.23 minutes). Furthermore, their total ridership volume nearly doubles from Wednesday to Saturday, indicating that their primary use case is weekend exploration rather than daily transport.
- **Trend 2: The Commuter Consistency**
- Annual members exhibit a "Utility Profile." Their behavior remains remarkably stable regardless of the day of the week, with trip durations hovering consistently around **12 minutes**. Their peak volume occurs Tuesday through Thursday during traditional "rush hour" windows, suggesting that the membership is primarily valued for its efficiency in daily Chicago transit.
- **Trend 3: The "Conversion Window" (Seasonality)**
- A massive growth trend was identified between May and June, where casual ridership increased by **58%**. This indicates a critical "activation window" for marketing. Since casual ridership peaks in August (323,545 rides) but drops by over 90% in January, the value proposition for an annual membership should focus on the cost-savings during this high-intensity summer period.

5. Share: Visualizations

- **Data Story:** The main takeaway is that **Annual Members are the engine of Cyclistic.** They aren't just commuting, they drive our massive summer peaks and are the primary users of our premium electric fleet. However, **casual riders** are "Weekend Warriors." They take much longer trips and flock to major leisure hubs, specifically **DuSable Lake Shore Dr & Monroe St**. The opportunity is clear these riders already use the system heavily during the summer. We don't need to change their habits, we just need to show them that a membership makes the rides they are *already* taking more affordable.

- **Visual Evidence:**



6. Act: Recommendations

- **Top 3 Recommendations:**
- **1. Launch a "Summer Peak" Membership Campaign**
- Since both members and casual riders hit their highest usage from **June to August**, this is our prime conversion window. We should launch a targeted marketing campaign during these months.
    a. **The Action:** Offer a "Summer-to-Annual" promotion where casual riders can sign up for a discounted annual membership if they join during the peak season.
    b. **The Why:** Casual riders are already using the bikes like members during these months; we just need to lower the barrier to entry while their interest is highest.
- **2. Implement Location-Based "On-Dock" Marketing**
- The data pinpointed specific stations—especially **DuSable Lake Shore Dr & Monroe St**—as the most popular locations for casual riders.
    a. **The Action:** Place physical marketing at these specific top-10 stations, such as QR codes on the docking stations that offer an instant discount on an annual membership.

    b. **The Why:** By advertising exactly where casual riders are already starting their leisure trips, we reach our target audience at the precise moment they are experiencing the value of the service.
- **3. Market the "Electric Fleet" as a Membership Perk**
- The data shows that annual members are high-frequency users of the **Electric bikes**. Casual riders are interested in them, but members have truly embraced them for their utility.
    a. **The Action:** Market annual memberships not just as a "cost-saver," but as a "VIP Upgrade" that includes reduced surcharges for electric bikes or exclusive "Members-Only" access to electric fleet reserves during high-traffic weekends.
    b. **The Why:** This creates a value proposition based on the "premium experience" that our most loyal users already prefer.